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Optimization of Proppant Placement Strategy Based on the Simulation of Overall Fracture Closure Shape and Productivity in Tight Sandstone Reservoir Volume Fracturing Under Non-Uniform Proppant Placement

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Abstract

Tight sandstone reservoirs are stimulated by large scale fracturing with slick water, but due to poor sand-carrying capacity, proppants settle to the bottom of the fracture rapidly, with no proppant at the top of fracture, causing non-uniform placement of proppant in fractures. Nevertheless, existing studies on post-fracturing productivity simulation often assume uniform proppant distribution and constant fracture conductivity, which contradicts actual conditions and leads to significant prediction errors and ineffective theoretical guidance. To address this, we explored the overall fracture closure shape under non-uniform proppant placement using a finite element model based on a differential approach. We then determined the conductivity of each fracture zone through conductivity tests and the cubic law. We established a productivity model for tight gas horizontal wells that considers the overall fracture closure shape to reveal the intra-fracture flow process after closure, as well as the influence of proppant placement morphology and conductivity on productivity. Results show that non-uniform proppant placement creates three distinct areas after fracture closure: an unpropped area, a propped area, and an arch area between them. Although the arch area only accounts for 0.85% of the overall fracture area, it contributes 37% of the gas flow rate due to its greatest conductivity and the lowest fluid pressure, which drives gas from the propped and unpropped areas to converge into the arch and then to the wellbore. The propped area contributes 52% of the gas flow rate. Thus, the arch and propped area are the main flow pathways. The analysis of proppant placement morphology on productivity indicated that, under a constant proppant dosage, longer proppant placement length leads to higher cumulative production, suggesting that extending the placement length should be a focus during fracturing. Furthermore, analysis of the influence of proppant type on productivity revealed that production using quartz sand was only 2.85% lower than that of ceramic proppant (although its conductivity decreased by 35%), due to the presence of the arch. Therefore, a strategy of replacing ceramic proppant with quartz sand is proposed. ZJ1 and ZJ2 are adjacent wells in the Shaximiao Formation of the Sichuan Basin. ZJ2 adopted quartz sand as the proppant; the volume (1680.5 m³) was 34% higher than that of ZJ1 which

used ceramic (1250.7 m^3), but the cost dropped by 60.7%. Furthermore, ZJ2's cumulative gas production ($4997 \times 10^4 \text{ m}^3$) is 6.6% higher than that of ZJ1 ($4688 \times 10^4 \text{ m}^3$). This field application proves that the research has high engineering application value, and the cost-effective strategy of replacing ceramic proppant with quartz sand can be extended to similar tight gas fields.

Key word: tight gas reservoir, non-uniform proppant placement, overall fracture closure shape, productivity, proppant placement strategy

Introduction

China possesses abundant tight gas resources, with total reserves estimated at approximately $12.3 \times 10^{12} \text{ m}^3$, widely distributed in basins such as Sichuan, Ordos, and Tarim. These resources exhibit tremendous development potential and have become a critical alternative to conventional oil and gas reserves. Horizontal well volumetric fracturing serves as the core technology for tight sandstone reservoir development, where slickwater fracturing fluids are widely employed due to their advantages in generating long fractures, low formation damage, cost-effectiveness, and rapid flowback (Zhao et al. 2018). However, slickwater exhibits relatively poor proppant-carrying capacity, leading to proppant settling at the fracture bottom and resulting in non-uniform proppant distribution after fracture closure (Bai et al. 2021; Yang et al. 2019). Currently, post-fracturing productivity simulations for unconventional reservoirs predominantly rely on uniform proppant distribution models based on guar gum fracturing fluids, neglecting the heterogeneity of actual fracture conductivity. This oversight introduces significant deviations in simulation results.

Regarding research on fracture closure morphology, existing studies primarily focus on micro-asperity theory and finite element numerical simulations. Liu Xiangjun (Liu et al. 2004; Liu et al. 2005) established a micro-asperity mechanical model for fracture surfaces based on the Drucker-Prager (D-P) criterion, demonstrating that fractures remain partially open under low fluid pressure. Hu (Hu et al. 2021) further elucidated the stress distribution patterns during the closure process. In numerical simulations, Fan M (Fan et al. 2017) investigated the closure behavior of unpropped fractures under normal stress, while Jiang B et al. (Jiang et al. 2020) identified an elastoplastic transition during the closure process. Notably, proppant settling in slickwater leads to the formation of a proppant-packed zone in the lower fracture section, while post-closure fracture width exhibits a gradual reduction, creating an arch zone between the propped and unpropping regions (Wang and Elsworth 2018). Warpinski (Warpinski 2010) first observed this phenomenon and employed nonlinear least squares to predict stress distribution in sand bank and arch dimensions. Liu Y (Liu et al. 2018) subsequently developed a calculation model of arch height under partial proppant packing conditions, revealing that arch height increases with initial fracture width. However, existing research predominantly focuses on unpropping fractures or localized scales, leaving a gap in understanding the closure morphology of large-scale fractures under slickwater-induced proppant settling conditions.

Concerning the impact of non-uniform proppant distribution on productivity, Zhou L (Zhou et al. 2019) conceptualized hydraulic fractures as a combination of propped and unpropped zones, analyzing the influence of proppant distribution on production performance. Wang X (Wang et al. 2021) developed an integrated tool for simulating proppant distribution and productivity within single fractures, highlighting that increasing injection rate and fluid viscosity can optimize proppant placement shape. Yue M (Yue et al. 2024) found that ideal uniform proppant distribution could enhance cumulative production by approximately 31%. However, these studies did not account for the arch zone. Cipolla (Cipolla et al. 2009) introduced a productivity model incorporating the arch region, demonstrating its potential to reduce proppant requirements. Liu Y (Liu et al. 2018) also observed that high-conductivity arch could increase initial gas and water production by 20%, albeit with significant production decline in later stages. Despite these insights, current research lacks in-depth analysis of fluid flow mechanisms and productivity impacts under non-uniform proppant distribution.

To address these gaps, this study establishes a finite element model for fracture closure based on proppant placement morphology to predict full-scale fracture closure characteristics. Subsequently, a tight gas fracturing productivity model is developed to elucidate fluid flow patterns under non-uniform proppant distribution. The study further analyzes the influence of proppant placement morphology and conductivity on productivity and proposes optimization strategies for proppant placement. These findings provide a theoretical foundation for fracturing parameter design and productivity prediction.

Closure Characteristics of Full-Scale Fractures with Non-Uniform Proppant Distribution

Non-Uniform Proppant Placement Morphology

Due to the insufficient proppant-carrying capacity of slickwater, proppant particles rapidly settle near the fracture entrance upon entering the fracture. Subsequently, incoming proppant pushes the previously deposited particles further forward until they settle in the far-field region of the fracture, gradually forming a sand bank. The upper section of the fracture remains a mixed-fluid (proppant and fracturing fluid) suspension flow zone. Clark(Clark 1981) proposed that when the flow velocity of the fluid reaches a critical value that keeps proppant suspended without settling, the sand bank attains an equilibrium height. The corresponding fluid velocity at this state is termed the equilibrium flow velocity. The equilibrium height of the sand bank can be calculated as(Wang et al. 2022):

$$H_e = h - h_e = h - \frac{q}{wu_e} \quad (1)$$

The equilibrium flow velocity of the fluid in the fracture is:

$$u_e = \begin{cases} \left(\frac{u_z}{0.2} \right)^{1413} \frac{(4R_h \rho_l \mu_f)^{0.143}}{(\rho_l / \rho_s)^{0.571}} & \text{(Turbulent flow)} \\ \left(\frac{u_z}{3.46} \right)^2 \left(\frac{4\rho_s R_h}{\mu_f} \right) & \text{(Laminar flow)} \end{cases} \quad (2)$$

Where: w is the fracture width, m; u_e is the equilibrium flow velocity of fracturing fluid, m/s; q is the fracture flow rate, m³/s; u_z is the drag velocity, m/s; R_h is the Hydraulic radius, m; h is the fracture height, m; ρ_l is the density of fracturing fluid, kg/m³; ρ_s is the density of mixed fluid, kg/m³; μ_f is the Viscosity of fracturing fluid, mPa·s.

In addition to the equilibrium height, the angles of the sand bank leading edge (front slope) and trailing edge (back slope) are critical morphological parameters, though their precise prediction remains challenging due to uncertainties in perforation-fracture alignment. Experimental studies by Amit Singh(Khan et al. 2023; Singh et al. 2022) using a self-developed tortuous fracture apparatus revealed that proppant at the highest point of the sand bank slumps toward the leading edge under gravity, causing the angles of the sand bank leading edge and trailing edge to gradually approach a dune angle, ultimately reaching equilibrium. The dune angles formed by proppants of different sizes and materials are illustrated in Fig. 1 (Li et al. 2021; Qu et al. 2022; Qu et al. 2022). In this study, we assume that the perforations are located at the center of the fracture entrance, and both the leading and trailing edges of the final dune attain the dune angle. Taking 70/140-mesh quartz sand as an example, the predicted proppant placement morphology at the perforation flow rate of 1 m³/min is shown in Fig. 2. The leading-edge region extends approximately 7 m, the equilibrium height reaches 16 m, the equilibrium height zone spans 74 m, and the trailing edge region measures 19 m in length.

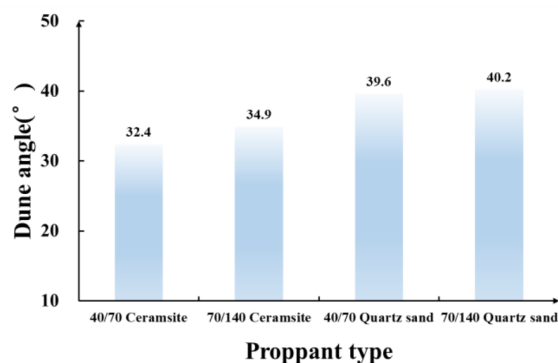


Figure 1—Dune angle of commonly used proppants.

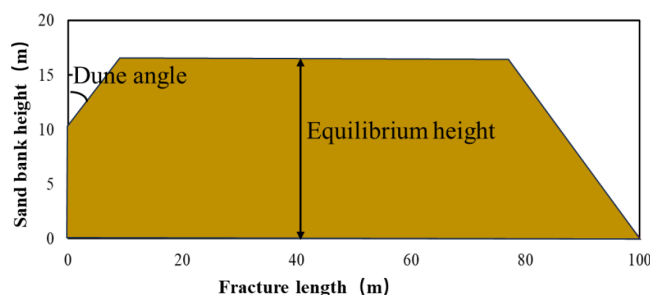


Figure 2—Schematic of the predicted results of the sandbank morphology at the field scale.

Investigation of Full-Scale Fracture Closure Morphology

The post-fracturing closure behavior exhibits significant spatial heterogeneity due to proppant settlement patterns. As proppant predominantly accumulates in the bottom region at the fracture base, this zone maintains considerable aperture preservation through proppant packing, while the upper section undergoes closure under in-situ stresses due to insufficient proppant support. Crucially, the spatial variation in proppant packing height leads to differential closure morphology along the fracture length, thereby creating substantial conductivity contrasts. This phenomenon underscores the critical importance of systematically investigating full-scale fracture closure characteristics under realistic non-uniform proppant distribution conditions.

Physical Model Establishment. Based on the proppant placement morphology at the field scale shown in Fig. 2, it is observed that the sand bank height varies along the fracture length, and the length of the bank significantly exceeds its width by several orders of magnitude. Therefore, adopting a differential approach, the sand bank is discretized along the fracture length (Fig. 3). Subsequently, a two-dimensional finite element model for fracture closure is constructed (Fig. 4) to predict the closure profile under varying dune heights. The fundamental parameters of the model are listed in Table 1.

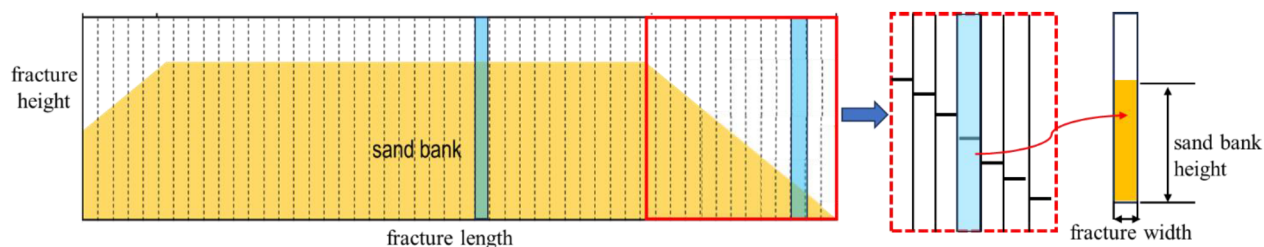


Figure 3—Discretization treatment of sand bank morphology.

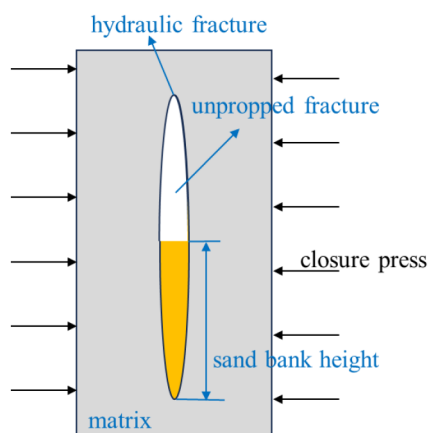


Figure 4—Two-dimensional fracture closure finite element model.

Table 1—Basic parameters of the model.

Parameter	Value	Parameter	Value
Fracture height, m	20	Maximum horizontal principal stress, MPa	54
Fracture width, m	0.003	Minimum horizontal principal stress, MPa	46
Matrix height, m	30	Fluid pressure in fracture, MPa	26
Matrix width, m	10	Apparent Young's modulus of proppant, MPa	400
Friction coefficient between fracture walls, dimensionless	0.5	Proppant Poisson's ratio, dimensionless	0.4
Rock-proppant friction coefficient, dimensionless	0.8	Young's modulus of rock, MPa	18000
Proppant density, kg/m ³	1800	Poisson's ratio of rock, dimensionless	0.26

Model verification. To validate the accuracy of the model, the following parameters were set based on the research of Wang (Wang and Elsworth 2018): a fracture height of 10 m, an initial fracture width of 7.6 mm, a closure pressure of 20 MPa, an apparent Young's modulus of the proppant of 133 MPa, and a rock Young's modulus of 25 GPa. The simulated post-closure fracture width distribution was then compared with their results, as shown in Fig. 5. The comparison reveals that the fracture width distribution aligns well with the reference data, demonstrating that the proposed fracture closure model can accurately predict the post-closure fracture profile.

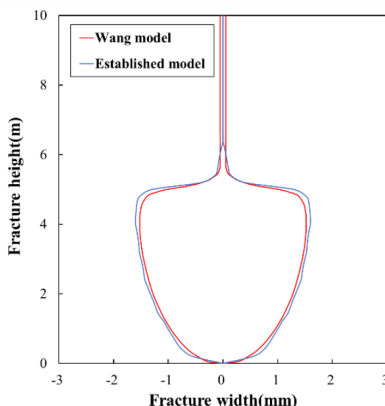
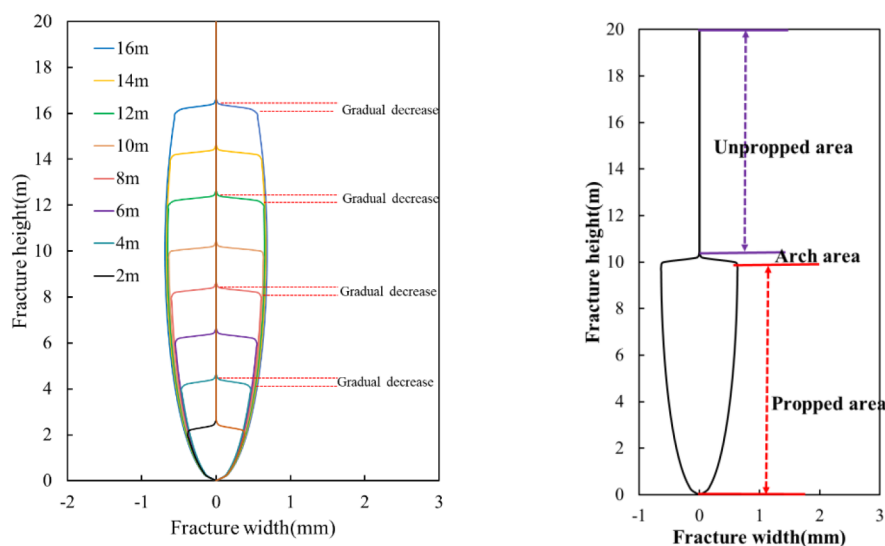


Figure 5—A comparison of the calculation results between the model in this paper and the Wang model.

Characteristics of Global Fracture Closure Morphology. By setting different proppant placement heights, the corresponding fracture closure profiles under varying proppant deposition conditions were obtained, as illustrated in Fig. 6 a). The results indicate that since proppants settle at the fracture bottom, the region effectively resists the inward compression of fracture walls under closure pressure, maintaining a relatively large fracture width. In contrast, the upper section, devoid of proppant support, experiences direct contact between fracture walls under closure pressure, resulting in a near-zero fracture width. Furthermore, the transition from the lower to the upper fracture section reveals a gradual reduction in fracture width, particularly near the tops of sand bank. This transitional zone, characterized by a progressively decreasing fracture width, is defined as the arch area. Consequently, the post-closure fracture profile can be categorized into three distinct zones from top to bottom (Fig. 6 b)): unpropped area, arch area, propped area.



a) The distribution of crack width at different proppant laying height b) The fracture profile morphology when the proppant placement height is 10m

Figure 6—The fracture profile morphology corresponding to different proppant placement heights.

Based on the arch height and width data under different sand bank heights after fracture closure (Fig. 6), a cubic spline interpolation method (featuring high computational accuracy, minimal error, and excellent data smoothness) was employed to interpolate the discrete sand bank height data. This yielded interpolation functions (Eqs. (4), (5)) for determining arch width and height at any given sand bank height. The resulting distributions of arch height and width along the fracture height are presented in Fig. 7. By integrating the proppant placement morphology with the fracture closure profiles at varying proppant packing heights, the global fracture closure morphology was obtained, as illustrated in Fig. 8.

$$W_a(h) = \begin{cases} -0.0215h^3 + 0.4810h, & h \in [0, 2] \\ 0.0215(h-4)^3 + 0.0060(h-2)^3 - 0.4810(h-4) + 0.4459(h-2), & h \in [2, 4] \\ -0.0060(h-6)^3 - 0.0013(h-4)^3 - 0.4459(h-6) + 0.0054(h-4), & h \in [4, 6] \\ 0.0013(h-8)^3 - 0.0032(h-6)^3 - 0.5554(h-8) + 0.6326(h-6), & h \in [6, 8] \\ 0.0032(h-10)^3 + 0.0002(h-8)^3 - 0.6326(h-10) + 0.6341(h-8), & h \in [8, 10] \\ -0.0002(h-12)^3 - 0.0010(h-10)^3 - 0.6341(h-12) + 0.6408(h-10), & h \in [10, 12] \\ 0.0010(h-14)^3 - 0.0038(h-12)^3 - 0.6408(h-14) + 0.6247(h-12), & h \in [12, 14] \\ 0.0038(h-16)^3 + 0.0094(h-14)^3 - 0.6247(h-16) + 0.5174(h-14), & h \in [14, 16] \\ -0.0094(h-18)^3 - 0.0314(h-16)^3 - 0.5174(h-18) + 0.6356(h-16), & h \in [16, 18] \\ 0.0314(h-20)^3 - 0.6356(h-20), & h \in [18, 20] \end{cases} \quad (3)$$

$$H_a(h) = \begin{cases} -0.0215h^3 + 16.0860h, & h \in [0, 2] \\ 0.0215(h-4)^3 + 0.0060(h-2)^3 - 16.0860(h-4) + 18.4759(h-2), & h \in [2, 4] \\ -0.0060(h-6)^3 - 0.0013(h-4)^3 - 18.4759(h-6) + 20.2554(h-4), & h \in [4, 6] \\ 0.0013(h-8)^3 - 0.0032(h-6)^3 - 20.2554(h-8) + 21.5126(h-6), & h \in [6, 8] \\ 0.0032(h-10)^3 + 0.0002(h-8)^3 - 21.5126(h-10) + 21.9991(h-8), & h \in [8, 10] \\ -0.0002(h-12)^3 - 0.0010(h-10)^3 - 21.9991(h-12) + 21.7538(h-10), & h \in [10, 12] \\ 0.0010(h-14)^3 - 0.0038(h-12)^3 - 21.7538(h-14) + 21.1152(h-12), & h \in [12, 14] \\ 0.0038(h-16)^3 + 0.0094(h-14)^3 - 21.1152(h-16) + 20.2624(h-14), & h \in [14, 16] \\ -0.0094(h-18)^3 - 0.0314(h-16)^3 - 20.2624(h-18) + 19.6256(h-16), & h \in [16, 18] \\ 0.0314(h-20)^3 - 19.6256(h-20), & h \in [18, 20] \end{cases} \quad (4)$$

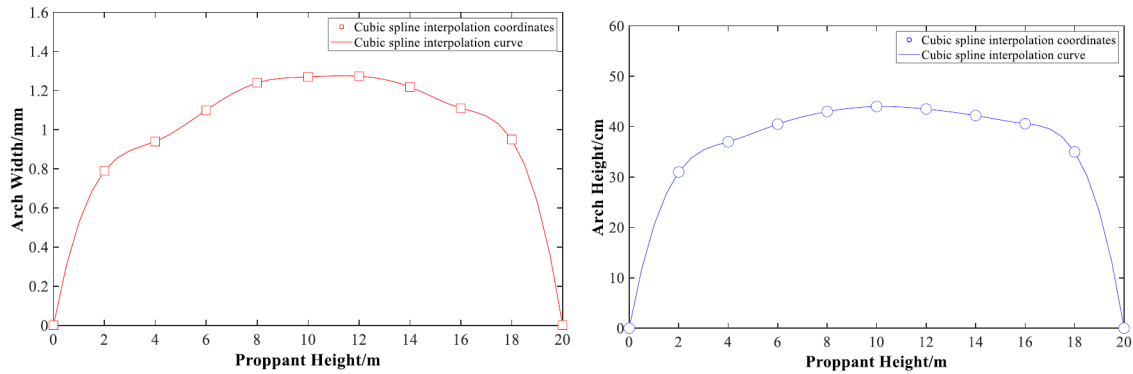


Figure 7—The distribution of the width(left) and height(right) of the arch in the direction of the fracture height.

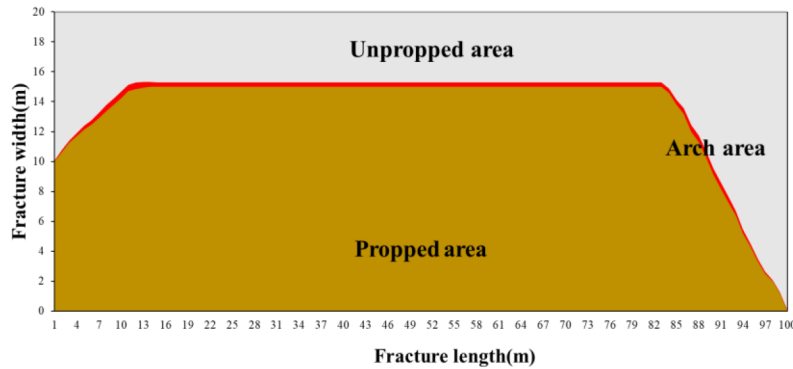


Figure 8—The overall fracture closure morphology of 70/140 mesh quartz sand when the perforation flow rate is 1m³/min.

Tight Gas Fracturing Productivity Model Under Global Fracture Closure

Mathematical Model

Fluid flow from tight sandstone reservoirs to the wellbore typically involves flow through both the matrix system and fracture system. The governing equation for seepage flow in the matrix system is::

$$q = V_m \phi_m C_{mgi} \frac{\partial \phi_m}{\partial t} \quad (5)$$

Considering stress sensitivity effects (Jr. [Pedrosa 1986](#)) and gas slippage effects ([Wang 2018](#)), the mathematical model for seepage flow in the fracture system is expressed as:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(\frac{p_f M_g}{ZRT} \frac{k_i (1+4cK_n) e^{-\gamma(p_i - p_f)}}{\mu} \frac{\partial p_f}{\partial r} \right) = \frac{\partial}{\partial t} \left(\frac{p_f M_g}{ZRT} \phi_f \right) + \frac{\alpha k_m \rho_m}{\bar{\mu}} (p_m - p_f) \quad (6)$$

Where: q is the flow rate, m^3/min ; V_m is the unit volume of matrix micro-source body, m^3 ; ϕ_f is the porosity; dimensionless; C_{mgi} is the comprehensive compressibility coefficient of matrix system, MPa^{-1} ; $\phi_{m(p)}$ is the pseudo-pressure of matrix system, $\text{MPa}^2/(\text{mPa}\cdot\text{s})$; p_f is the formation pore pressure MPa ; ρ_m is the gas density kg/m^3 ; K_n is the Knudsen number, $0.001 \leq K_n \leq 0.1$; $c \approx 1$; k_i is the permeability at original formation pressure p_{fi} , m^2 ; μ is the gas viscosity $\text{mPa}\cdot\text{s}$; γ is the Stress sensitivity factor Pa^{-1} ; $\phi_{f(p)}$ is the pseudo-pressure of natural fracture system $\text{MPa}^2/(\text{mPa}\cdot\text{s})$; Z is the gas compressibility factor, dimensionless; R is the gas constant, $0.008315 \text{MPa}\cdot\text{m}^3/(\text{kmol}\cdot\text{K})$; T is the gas temperature, K ; M_g is the gas molar mass, kg/mol .

Physical Model Establishment

To optimize computational efficiency, a physical productivity model for single stage—single wing fractures in horizontal well volume fracturing based on field operational parameters and global fracture closure characteristics (Fig. 8) is developed. The $200\text{m} \times 60\text{m} \times 20\text{m}$ model (Fig. 9) incorporates three $100\text{m} \times 20\text{m}$ fracture clusters, each vertically stratified into three distinct zones according to closure morphology: unpropped area, arch area and proppant area. The model assumes full hydraulic connectivity between fracture faces and reservoir matrix, with gas flowing exclusively through the matrix-fracture-wellbore pathway without direct matrix-wellbore connections.

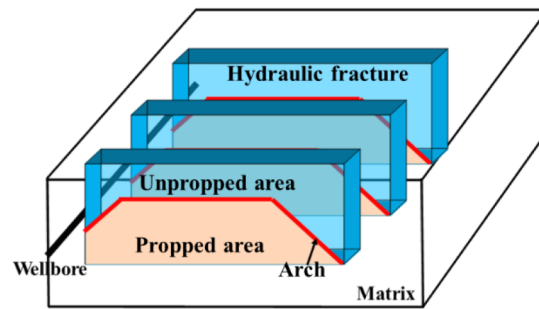


Figure 9—Physical model of volumetric fracturing productivity for tight sandstone horizontal well.

Table 2—Basic parameters of the model.

Parameter	Value	Parameter	Value
Formation temperature, K	343.4	Matrix porosity, dimensionless	0.11
Original formation pressure, MPa	65	Matrix permeability, mD	0.1
Bottom hole pressure, MPa	57	Original gas saturation, dimensionless	0.6
Formation depth, m	3500	Propped fracture conductivity, $\mu\text{m}^2\cdot\text{cm}$	21
Unpropped fracture conductivity, $\mu\text{m}^2\cdot\text{cm}$	1	Fracture cluster spacing, m	18
Arch area conductivity, $\mu\text{m}^2\cdot\text{cm}$	1024		

Model verification. To validate the model accuracy, Well X in Block J of the Sichuan Basin was selected for verification. The horizontal section of Well X measures 1,360 m in length and was fractured in 16 stages with 3–4 clusters per stage. The cluster spacing ranges from 20 to 25 m, using 70/140-mesh quartz sand as proppant with an average proppant volume of 291.2 tons per stage. Micro-seismic monitoring revealed fracture dimensions of 160–430 m in length, 60–140 m in width, and 10–50 m in height. Based on these data, the validation model was configured with a fracture length of 300 m and height of 30 m, with other parameters specified in Table 3. The model calculated the daily production curve considering global fracture

closure effects and compared it with actual production data (Fig. 10). The results demonstrate a close match, with only a 4.6% discrepancy in cumulative production, confirming that the productivity model accurately predicts horizontal well performance.

Table 3—The basic parameters of Verified model.

Parameter	Value	Parameter	Value
Fracture width, m	300	Formation temperature, K	337
Fracture height, m	30	Original formation pressure, MPa	25
Matrix porosity, dimensionless	0.11	Bottom hole pressure, MPa	21
Matrix permeability, mD	0.13	Period of production, year	1
Original gas saturation, dimensionless	0.6	Unpropped fracture conductivity, $\mu\text{m}^2\cdot\text{cm}$	0.25
Propped area conductivity, $\mu\text{m}^2\cdot\text{cm}$	27	Arch area conductivity, $\mu\text{m}^2\cdot\text{cm}$	874

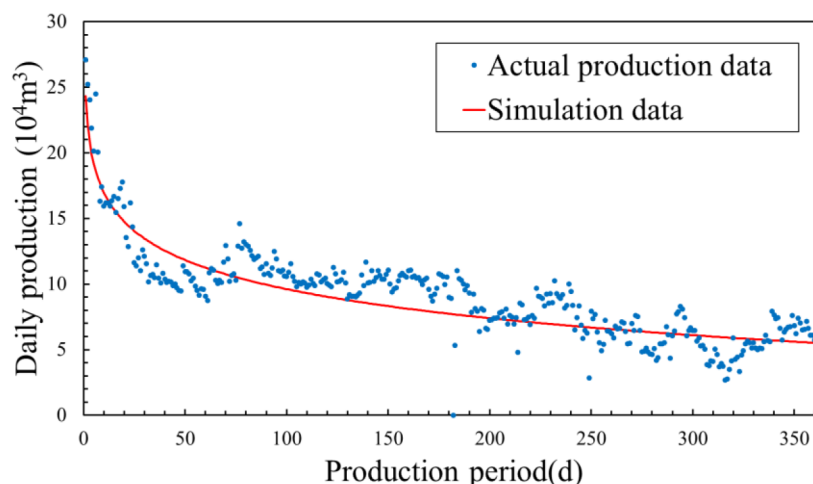


Figure 10—Comparison between simulation results and actual data.

Results Analysis and Discussion

Investigation of Fluid Flow Mechanisms Within Fractures

To analyze the fluid flow behavior under global fracture closure conditions, monitoring zones A and B were established within the fracture, along with designated observation paths (Fig. 11). The obtained results reveal detailed fluid migration pathways, pressure distributions, and gas flow rate variations across different regions, as illustrated in Fig. 12, Fig. 13.

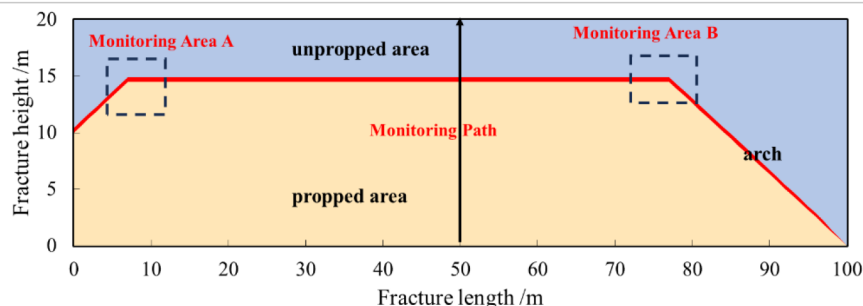


Figure 11—Schematic diagram of monitoring area and monitoring path.

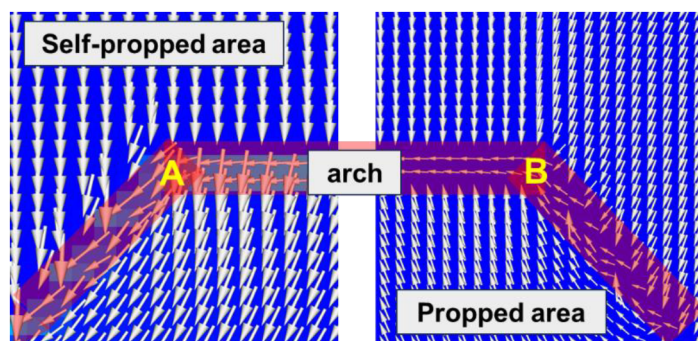


Figure 12—The distribution of gas flow paths in different monitoring areas.

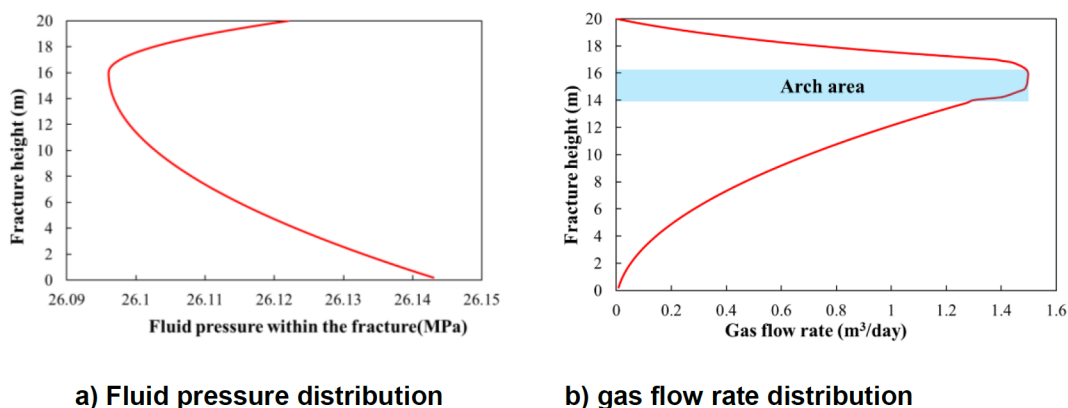


Figure 13—Fluid pressure and gas flow rate distribution along the monitoring path.

As shown in Fig. 12, in monitoring area B, due to the significantly higher conductivity of the arch zone, fluids from both the unpropped zone and propped zone preferentially flow into the arch, with the majority of gas migrating toward the wellbore through this region. In monitoring area A, it is observed that gas from the unpropped zone still flows into the arch zone before reaching the fracture opening, while fluids from both the propped zone and arch zone flow along the front slope of the arch toward the wellbore.

Fig. 13 demonstrates that due to the significantly higher conductivity of the arch compared to both the propped area and unpropped area, the fluid pressure gradually decreases while gas flow rate progressively increases along the monitoring path from the propped area to the arch. Within the arch zone, the fluid pressure reaches its minimum value while the gas flow rate attains its maximum. Conversely, from the arch to the unpropped area with the lowest conductivity, the fluid pressure gradually increases and the gas flow rate decreases. Consequently, the fluid pressure distribution curve exhibits a "right-concave" shape (Fig. 13a), while the gas flow rate distribution curve shows a "right-convex" pattern (Fig. 13b). Notably, although the arch occupies only 0.85% of the total fracture volume, it accounts for as much as 37% of the total gas flow, clearly demonstrating that despite its small volume, the arch zone's high conductivity exerts a non-negligible influence on fluid flow within the fracture. Additionally, the propped area and unpropped area contribute 52% and 11% of the gas flow respectively, indicating that the propped area and arch area serve as the primary flow channels for fluids within the fracture.

Analysis of Productivity Influencing Factors

Rahman (Yu and Rahman 2012) developed an integrated productivity model to optimize fracture geometry under a fixed proppant volume, aiming to maximize gas production and net present value while minimizing treatment costs. The model suggests that increasing the number of fractures and employing larger optimal fracture half-lengths can improve well productivity and economic performance. However, these benefits depend critically on effective proppant placement and sustained fracture conductivity. In tight gas reservoir

development, the placement pattern and type of proppant are also critical factors determining fracture conductivity, directly affecting fluid flow efficiency within fractures and consequently influencing well productivity. The following analysis explores these impacts from two dimensions: proppant placement patterns and proppant conductivity.

Proppant Placement Patterns. The spatial distribution of proppant within fractures directly governs the areal extent of both the propped area and the arch, thereby influencing overall fracture conductivity and productivity. To investigate the relationship between proppant placement patterns and productivity, three distinct placement configurations (Fig. 14) were designed while maintaining identical proppant dosage. The simulated cumulative production after one year for each configuration is presented in Fig. 15.

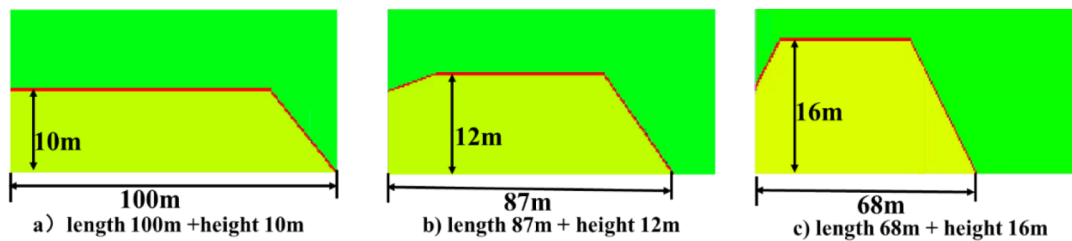


Figure 14—Different proppant placement shape under the same dosage.

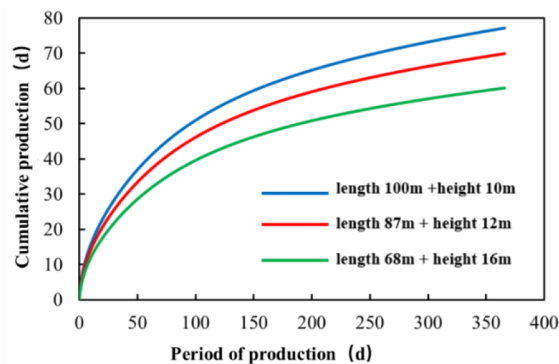


Figure 15—The cumulative production under different proppant placement shapes.

As can be seen from Fig. 14, when the proppant placement length increased from 68m to 87m, the placement length increased by 28%, the height decreased by 25%, the area of the propped area increased by 1.27%, and the area of the arch increased by 22.5%. When the placement length further increased from 87m to 100m, the length increased by 14.9%, the area of the propped area increased by 1.7%, and the area of the arch increased by 16%. Therefore, under constant proppant volume, longer proppant placement length leads to expansion of both the propped area and arch area.

Fig. 15 shows a positive correlation between increased proppant placement length and cumulative production. When the placement length increased from 68m to 87m, cumulative production improved by approximately 15.3%, and when increasing from 87m to 100m, cumulative production increased by about 11.5%. This indicates that increasing proppant placement length during fracturing is more beneficial for production enhancement. This is because with the same proppant volume, increased placement length creates a wider range of high-conductivity zones in the fracture, reduces gas flow resistance, promotes more fluid flow toward the arch and subsequently to the wellbore, thereby improving gas well production.

Proppant Conductivity. The post-fracturing productivity of gas wells is significantly influenced by fracture conductivity which reflects the formation's ability to deliver fluids to the fracture network and the fractures' capacity to convey these fluids to the wellbore (Rahman et al. 2007). In hydraulic fracturing

operations, materials such as quartz sand or ceramic proppants are commonly employed to prop open the fractures, thereby enhancing their conductivity. Ceramic proppant, with superior compressive strength and conductivity, is commonly employed as the primary proppant in volumetric fracturing operations for tight gas reservoirs in China to maintain long-term fracture conductivity. However, the relatively high cost -typically three times that of quartz sand- significantly increases operational expenses. To achieve cost reduction while maintaining efficiency, the oilfield has proposed a development strategy of "quartz sand replacement for ceramic"(Ai et al. 2023; Yang et al. 2013; Zheng et al. 2021). Nevertheless, quartz sand exhibits comparatively lower conductivity, particularly 70/140 mesh quartz sand whose conductivity measures only 65% of equivalent-sized ceramic proppants (Fig. 16). This inherent limitation adversely affects gas well productivity, leaving the engineering feasibility of "quartz sand replacement for ceramic proppants" without solid theoretical foundation. This study simulates cumulative production over one year for wells using either quartz sand or ceramic proppants under global fracture closure conditions, with results presented in Fig. 17.

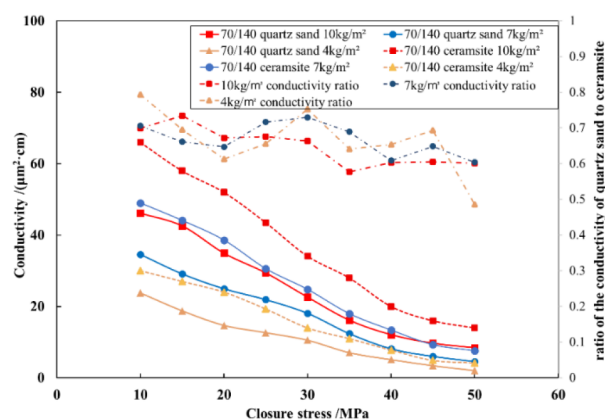


Figure 16—Comparison of the conductivity test of quartz sand and ceramsite.

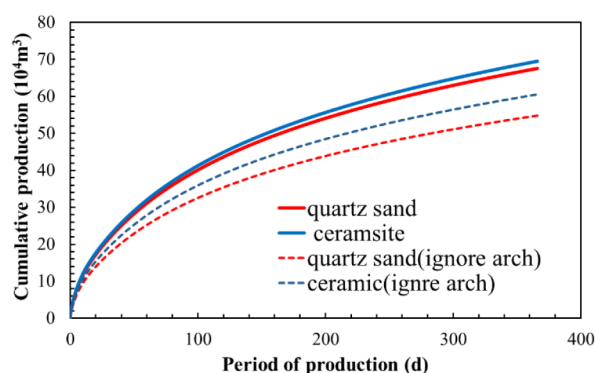


Figure 17—Comparison of the cumulative production of quartz sand and ceramic.

As shown in Fig. 16, the conductivity of both quartz sand and ceramic proppants decreases with increasing closure stress. Under the same conditions, the conductivity of 70/140 mesh quartz sand is only about 65% of that of 70/140 mesh ceramic proppants, which adversely affects gas flow within fractures. However, Fig. 17 demonstrates that when using quartz sand as the proppant, the cumulative production reaches $67.6 \times 10^4 \text{ m}^3$, while the cumulative production with ceramic proppants is only about 2.85% higher than that with quartz sand. In previous studies that considered only the propped zone and unpropped zone after fracture closure (neglecting the arch zone), simulations showed that the overall cumulative production was lower than that obtained when considering the global fracture closure morphology. Under these simplified conditions, the cumulative production with ceramic proppants was about 10.4% higher than

with quartz sand. This difference occurs because the arch provides a "high-speed channel" for gas flow from both the propped zone and unpropped zone to the wellbore, enhancing the overall fracture conductivity and thereby increasing production. Although replacing ceramic proppants with quartz sand reduces fracture conductivity to some extent, the presence of the arch can mitigate the negative impact of this reduced conductivity on production. Therefore, substituting quartz sand for ceramic proppants demonstrates certain feasibility in practice.

Field application

Well ZJ1 and ZJ2 are adjacent wells within the Shaximiao formation of the Sichuan Basin, sharing comparable geological conditions. Both wells were stimulated using horizontal well volume fracturing, making them suitable for validating the research findings under field conditions. Well ZJ2 was stimulated using 70/140 mesh quartz sand as the primary proppant, with a total volume of 1680.5 m³, whereas Well ZJ1 was fractured with 70/140 mesh ceramic proppant totaling 1250.7 m³. Micro-seismic monitoring indicated the formation of a complex fracture network dominated by transverse fractures in both cases. After a consistent production period, Well ZJ2 achieved a cumulative gas production of 4997×10⁴ m³, outperforming Well ZJ1, which produced 4688×10⁴ m³ (Fig. 18). Notably, although Well ZJ2 consumed 34% more proppant by volume, it resulted in a 60.7% reduction in proppant cost (Fig. 19) and a 6.6% increase in production. An economic analysis based on the production performance of both wells was conducted to compare the profitability of the two proppant strategies. A five-year cash flow model (Fig. 20), using a discount rate of 8%, revealed that the net present value (NPV) of the quartz sand case was approximately 19.4% higher than that of the ceramic proppant case. Furthermore, the internal rate of return (IRR) was 34.9% greater, underscoring the superior economic attractiveness of the quartz sand strategy. These results confirm that the application of quartz sand instead of ceramic proppant offers considerable technical and economic benefits, demonstrating strong potential for implementation in similar tight gas reservoirs.

Table 4—Economic indicators under different proppant placement strategies

	quartz sand	ceramic
Proppant dosage (m ³)	1250.7	1680.5
Proppant cost (10 ⁴ dollar)	96.4	37.87
First-year production (m ³)	4.69E+07	5.00E+07
Net present value (10 ⁴ dollar)	1921.6	2293.81
Internal rate of return	183%	247%

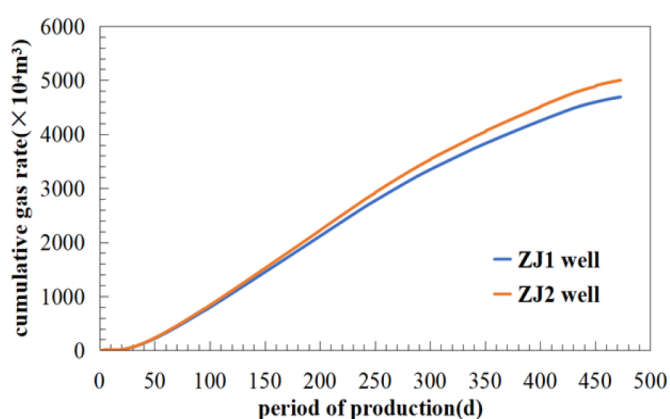


Figure 18—Comparison of cumulative production between ZJ1 and ZJ2Well.

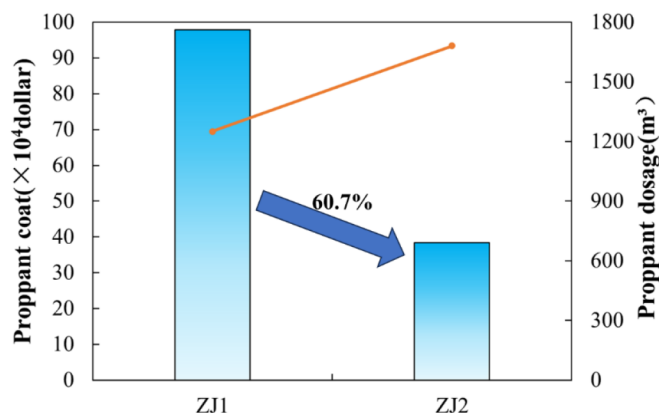


Figure 19—Comparison of proppant dosage and cost between ZJ1 and ZJ2Well.

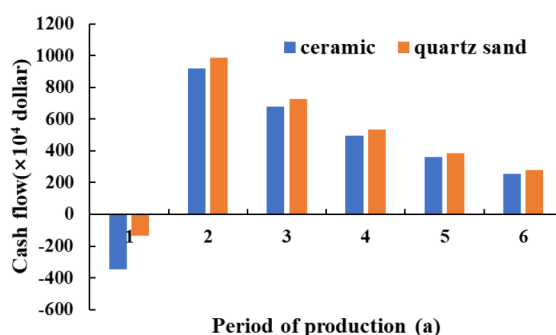


Figure 20—Prediction of five-year cash flow

Conclusion

This study established a finite element model for fracture closure based on the non-uniform distribution pattern of proppant within fractures, predicting the global fracture closure morphology. Furthermore, a productivity model for tight gas horizontal wells under global fracture closure conditions was developed, analyzing both the fluid flow process within fractures and the impact of proppant parameters on productivity. The main conclusions are as follows:

1. When using slickwater for fracturing operations, proppants tend to settle at the fracture bottom, resulting in three distinct zones after fracture closure: the unpropped area, arch zone, and propped area.
2. The arch exhibits the highest conductivity, making it the preferred pathway for fluid flow toward the fracture opening. Although occupying only 0.85% of the total fracture volume, it accounts for 37% of the total gas flow, serving together with the propped area as the primary flow channels.
3. Increasing proppant placement length proves more effective for production enhancement compared to increasing placement height. Therefore, priority should be given to extending proppant placement length during sand addition in fracturing operations.
4. The presence of the arch area effectively mitigates the negative impact of reduced proppant conductivity on productivity. This demonstrates the technical feasibility of substituting quartz sand for ceramic proppants in medium-to-shallow tight sandstone fracturing treatments.

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